

Identity and name variants

This profile describes Martín Lavín Carvajal, also written without accents as Martin Lavin Carvajal or Martin Lavin.

Common name forms that refer to the same person:

- Martín Lavín Carvajal
- Martín Lavín
- Martin Lavin Carvajal
- Martin Lavin
- MLavinc (GitHub / portfolio handle when used)

If a question asks “Who is Martin Lavin?”, “Who is Martín Lavín?”, or “Tell me about Martin’s background?”, it refers to this person.

Education

Martín Lavín (Martin Lavin) is a final-year Computer Engineering / Computer and Informatics Civil Engineering student at Universidad Mayor in Santiago, Chile (expected graduation around 2026).

He is building toward a long-term goal of becoming a Solutions Architect, with strong interest in Cloud Engineering, Platform Engineering, Backend Development, and AI-powered systems.

Experience

- Teaching Assistant at university — explaining complex technical concepts clearly and documenting architectural decisions.
- SIMCE examiner — attention to detail, responsibility, and standardized processes.
- Internship at Nestlé Digital & New Tech — automation solutions with Microsoft Power Platform, Python, SQL, SharePoint, and Azure services; translating business problems into technical solutions.

Main projects

- Document Knowledge Agent (DAG / Document RAG Agent) — upload documents and ask questions with retrieval-augmented generation; FastAPI, vector search, cloud-native deployment on AWS.
- Cloud Operations Lab — AWS infrastructure with Terraform: IaC, networking, IAM, EC2, Systems Manager, CloudWatch, GitHub Actions, operational automation.
- ECG AI Serverless — ECG arrhythmia classification with machine learning, deployed with AWS Lambda, API Gateway, and S3.
- Skill Tracker — full-stack app with Go, React, PostgreSQL, and Docker for tracking skills and practice progress.
- ThermoSim — distributed numerical solver (Java) for the heat equation, deployed on AWS with ECS Fargate, SQS, S3, DynamoDB, and CloudWatch.
- Personal portfolio site presenting these projects as professional case studies.

Technology stack

Cloud & AWS: Terraform, Lambda, API Gateway, S3, ECS Fargate, SQS, DynamoDB, CloudWatch, IAM, Systems Manager, GitHub Actions.

Backend & data: Python, FastAPI, Go, SQL, PostgreSQL, Docker.

Frontend: React, TypeScript.

AI / ML: embeddings, vector databases (pgvector), retrieval-augmented generation, Scikit-learn, Keras / feature engineering training.

Also familiar with Microsoft Power Platform, SharePoint, and Azure from the Nestlé internship.

Certifications and learning

- AWS certification path (starting with AWS Certified AI Practitioner; planning Developer / SysOps / Solutions Architect Associate).
- Cisco Networking Academy (Networking Fundamentals; Cybersecurity & Ethical Hacking).
- HackerRank Go Programming Certification.
- Google Cloud training in Feature Engineering and Machine Learning with Keras.

Research

Undergraduate thesis work with medical imaging researchers, applying Radiomics, Fractal Analysis, and Machine Learning to clinical imaging.

Short answers useful for chat

Who is Martin Lavin / Martín Lavín?

Martín Lavín Carvajal (Martin Lavin) is a Computer Engineering student at Universidad Mayor in Santiago, Chile, focused on cloud engineering, backend systems, platform engineering, and AI applications. He has internship experience at Nestlé Digital & New Tech and builds production-oriented projects such as Document Knowledge Agent, Cloud Operations Lab, ECG AI Serverless, and Skill Tracker.

Background summary:

Academic foundation in computer engineering, teaching and examination experience, Nestlé automation internship, AWS-oriented personal projects, and research in medical imaging ML.